

```
// Quit when all windows are closed.
app.on('window-all-closed', () => {
  // On macOS it is common for applications and their menu bar
  // to stay active until the user quits explicitly with Cmd + Q
  if (process.platform !== 'darwin') {
    app.quit()
  }
})
app.on('activate', () => {
  // On macOS it's common to re-create a window in the app when
  // the dock icon is clicked and there are no other windows
  open.
```

```
if (win === null) {   createWindow()
}
})
```

The `index.html` would consist of the source code of the web page. A sample is shown below:

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="UTF-8">
    <title>Hello World!</title>
  </head>
  <body>
    <h1>Hello World!</h1>
    We are using node
  <script> document.write(process.versions.node)</script>,
  Chrome <script> document.write(process.versions.chrome)</
  script>, and Electron <script> document.write(process.
  versions.electron)</script>.
  </body>
</html>
```

Now, we have built all the three required files: `main.js`, `package.json` and `index.html`. These will be executed by running the `electron` command in the source directory (if Electron is installed globally with npm).

electron

Packaging Electron apps

Once the development of the application is over, it can be packaged for distribution. Detailed package building instructions for manually creating the packages are given at <https://electron.atom.io/docs/tutorial/application-distribution/>. Apart from the manual packaging, there are specific tools available to assist in the package building process. Some of these are listed below:

- Electron-forge
- Electron-builder
- Electron-packager

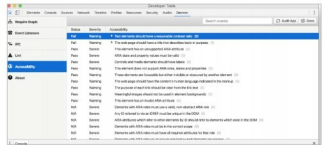


Figure 4: Devtron – accessibility analysis

Accessibility testing of apps

Accessibility of applications has become an important factor these days. The Electron apps can be made accessible effectively and this aspect can be tested. Though the GUIs in the Electron apps are HTML pages, their accessibility cannot be audited directly using accessibility audit tools, as there is no direct URL available for these pages. To facilitate accessibility auditing of the Electron apps, new functionalities have been introduced to Devtron (an Electron DevTools extension—more information can be got at <https://electron.atom.io/devtron/>) and Spectron (an Electron testing framework - <https://electron.atom.io/spectron/>).

A sample code for accessibility auditing with Spectron is shown below:

```
app.client.auditAccessibility().then(function (audit)
{
  if (audit.failed)
  {
    console.error(audit.message)
  }
})
```

Similarly, the accessibility tab has been introduced in Devtron too. A sample screenshot is shown in Figure 4.

This article is a basic introduction to the Electron framework. If you are interested in further exploring it, there are a large number of resources available. The community (<https://electron.atom.io/community/>) Web page of the Electron framework offers a lot to those interested in getting detailed insights. **END**

References

- [1] Electron: <https://electron.atom.io/>
- [2] <https://electron.atom.io/docs/tutorial/quick-start/>
- [3] <https://electron.atom.io/docs/>

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